Nepal College of Information Technology

Assessment- Spring- 2026

Program: BE-CE
Time: 3 Hours
Semester: VI
FM: 100
Subject: Simulation and Modeling
PM: 45

- ✓ *Candidates are requested to give their answer as far as practicable in their own words.*
- ✓ *The figure in the margin indicates the full marks*
- ✓ ***Attempt ALL question***

1) a) What is a model? What are the various types of model that represents a system? 7

b) Describe the Monte Carlo method and use it to get the value of pi. 8

2) a) Differentiate system simulation and real time simulation with example. 7

b) What is differential equation? How they can be used to represent continuous system? Explain with example. 8

3) a) What is CSSLs? Describe the structure of CSMP with reference of an example. 7

b) Calls get lost when cannot be connected at the time of arrival. Show the various states to represent the system. 8

4) a) What do you understand by simulation programming task? Explain in detail. 7

b) When the output obtained is run multiple times which of the analysis method is appropriate? Explain. 8

5) a) Consider the following 60 values and test whether 2nd, 9th, 16th,..... numbers in the sequence are auto correlated, where $\alpha=0.05$.

0.30	0.48	0.36	0.01	0.54	0.34	0.96	0.06	0.61	0.85
0.48	0.86	0.14	0.86	0.89	0.37	0.49	0.60	0.04	0.83
0.42	0.83	0.37	0.21	0.90	0.89	0.91	0.79	0.57	0.99
0.95	0.27	0.41	0.81	0.96	0.31	0.09	0.06	0.23	0.77
0.73	0.47	0.13	0.55	0.11	0.75	0.36	0.25	0.23	0.72
0.60	0.84	0.70	0.30	0.26	0.38	0.05	0.19	0.73	0.44

8

compiled by Niraj Bhatta

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b) The sequence of numbers 0.54, 0.73, 0.98, 0.11 and 0.68 has been generated. Use KS test with α=0.05 to determine if the hypothesis that the numbers are uniformly distributed on the interval [0, 1] can be rejected. 7

6) a) What is preamble section in SIMSCRIPT? How are variables, entities and events declared in SIMSCRIPT? 7

b) Parts are being made at the rate of one every 6 minutes. They are of two types A and B and are mixed randomly with about 10% being type B. A separate inspector is assigned to examine each type of part. The inspection of A parts takes 4±2 minutes and B parts takes 20±10 minutes. Both inspectors reject about 10% of the parts they inspect. Simulate for a total of 1000 type A parts accepted. 8

7) Write Short notes on (any two): 10
   a) Characteristics of queue
   b) Feedback system
   c) Elimination of initial Bias

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2083/04/14			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	VI	1.5 hrs	

Subject: - Simulation and Modeling

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt <u>All</u> questions.*
- ✓ *The figures in the margin indicate <u>Full Marks</u>.*
- ✓ *Assume suitable data if necessary.*

1. a The following numbers have been generated 0.54, 0.73, 0.98, 0.11 and 0.68. Use KS test to check whether given numbers are uniformity distributed or not at 95% confidence interval. [7]

   b) Using chi-square test with α = 0.05, Check for the uniformity of the given random no. [8]

0.87	0.69	0.12	0.83	0.60	0.95	0.01
0.93	0.27	0.58	0.23	0.99	0.75	0.19
0.28	0.15	0.88	0.36	0.89	0.33	0.68
0.31	0.35	0.49	0.64	0.91	0.05	0.28
0.41	0.43					

2. a) Write GPSS program and draw the block diagram for anufacturing shop model when only one inspection is possible at a time with necessary assumption. [8]

   b) Explain the organization of SIMSCRIPT program with suitable diagram

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3. a) What is CSSL and why were they developed? Explain the structure of CSMP III code for automobile Suspension system [8]

   b) What are the various types of calls? Simulate telephone system for lost and delayed call system. [7]

4. Write short notes on: (***Any one***) [1*5=5]
   a) Elimination of Initial bias
   b) Utilization and Occupancy

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2083/03/5			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	VI	1.5 hrs	

Subject: - Simulation & Modeling

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt <u>All</u> questions.*
- ✓ *The figures in the margin indicate <u>Full Marks</u>.*
- ✓ *Assume suitable data if necessary.*

1. a) Describe in brief the different types of simulation models. Taking an example of any realistic simulation problem, illustrate the concepts of system boundary, environment, entities, endogenous and exogenous activities. [8]
   b) Solve the following using Monte Carlo Simulation. Use 15 iterations to generate the number. Also, estimate the error percentage. [7]
   $I = \int 0 \text{ to } \frac{\pi}{2} \sin x \cos x \, dx.$
2. a) Compare simulation and analytical methods with example. [8]
   b) Define differential equation. Explain why it is required in continuous system Simulation. Explain the types of differential equation with example. [7]
3. a Calculate different parameter for single server queuing system when arrival time and service times are given below: [8]

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Customer	Arrival time	Service time
1	0	3
2	3	2
3	8	4
4	2	3
5	9	2
6	10	5
7	16	6

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I Average waiting time for customers.
ii. Probability that a customer has to wait in the queue.
iii. Average time customer spends in the system.

4. a) What is the significance of incorporating delayed calls in a telephone system simulation? Provide a detailed description of the simulation involving these delayed calls and how they impact the overall system performance. [7]

   b) Draw cobweb model for market economy for following equation and fluctuation of market price
   $D=12.4-1.2P$
   $S=8.0-0.6P_{-1}$
   $D=S$ and $P_0=1.0$ [8]

5. Write short notes on (Any one): [1×5 = 5]
   a) Validation and Verification
   b) Queuing system

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Nepal Engineering College

Level: Bachelor
Programme: BE
Course: Simulation and Modeling

Semester: Spring
Year : 2026
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is a model? Explain different types of models. 7
   b) Define simulation with appropriate examples. What are the steps to be followed in simulation? Explain with a neat diagram. 8
2. a) What are the advantages and disadvantages of the Monte Carlo method? 8
   Estimate the value of integral of sin(x) from 0 to 180 degree interval.
   b) What is the queue? What are the measures of the queuing system? Explain with reference to Single Server Queuing System. 7
3. a) What is an analog method? Design an analog computer for the human liver. 7
   The equations for human liver are
   $dx_1/dt = -k_{12}\, x_1 + k_{21}\, x_2$
   $dx_2/dt = k_{12}\, x_1 - (k_{21} + k_{23})\, x_2$
   $dx_3/dt = k_{23}\, x_2$
   b) Explain about the simulation of an automobile wheel with CSMP III program for it.

   Or

   Explain about the simulation of an auto pilot aircraft system with CSMP III program for it.
4. a) What is discrete system simulation? Explain with reference to lost and delay call telephone system simulation.
   b) How are arrival patterns generated in simulation? Explain with suitable examples.
5. a) What are the two desired properties of a random number? For the following sample of random numbers, perform the test for independence using the K-S test. ($D_{0.05,10} = 0.41$)
   0.35, 0.77, 0.12, 0.33, 0.88, 0.45, 0.19, 0.25, 0.91, 0.54
   b) A sequence of 1000 four digit numbers has been generated and analysis indicates the following combinations with frequencies:

Combination (i)	Observed Frequency {Oi }
Four different digits	570
One pair	380
Two pairs	34
Three like digits	15
Four like digits	1

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Based on the poker test, check whether the numbers are independent. Use c=0.10 and N=4 is 7.78

Or

Based on the frequency with which gaps occur, analyze the following 110 digits to test whether they are independent. Use level of significance 5%. [D0.05,100=0.136]

4, 1, 3, 5, 1, 7, 2, 8, 2, 0, 7, 9, 1, 3, 5, 2, 7, 9, 4, 1, 6, 3 ,3, 9, 6, 3, 4, 8, 2, 3, 1, 9, 4, 4, 6, 8, 4, 1, 3, 8, 9, 5, 5, 7, 3, 9, 5, 9, 8, 5, 3, 2, 2, 3, 7, 4, 7, 0, 3, 6, 3, 5, 9, 9, 5, 5, 5, 0, 4, 6, 8, 0, 4, 7, 0, 3, 3, 0, 9, 5, 7, 9, 5, 1, 6, 6, 3, 8 ,8, 8, 9, 2, 9, 1, 8, 5, 4, 4, 5, 0, 2, 3, 9, 7, 1, 2, 0, 3, 6, 3 (110 numbers)

6. a) What are the two permanent entities of GPSS? Consider a factory that manufactures the football taking 20 to 40 minutes. The ball is moved from the generation to the inspection machine taking 2 minutes. There are 3 inspection machines at one place and need 30 to 60 minutes for inspection and reject 30% of the football. Draw a GPSS diagram to simulate the system. 8

b) How is output analysis done in estimation method? Explain with necessary expressions. 7

7. Write short notes on: (Any two) 2×5

a) Cobweb model
b) Measuring Utilization and Occupancy
c) Pure Pursuit Model
d) Organization of SIMSCRIPT

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Madan Bhandari College of Engineering

Urlabari-3, Morang

Final Internal Examination (Spring 2026)

Level: Bachelor — Full Marks: 60

Programme: B. E. Computer — Pass Marks: 30

Year/Part: III/II — Time: 2 hrs

Subject: - Simulation and Modeling

Attempt any 7 questions (From question 1 - 8) **7*8 = 56**

1. Define and develop a Poker test for four-digit random numbers. A sequence of 10000 random numbers each of four digits has been generated. The analysis of the numbers reveals that in 5120 numbers all four digits are different,4230 contain exactly one pair of like digits,560 contain two pairs,75 have three digits of a kind and 15 contain all like digits. Use Poker test to determine whether these numbers are independent? (Critical value of chi-square for α=0.05 and N=4 is 9.49).

2. Why do we need Monte Carlo method of simulation ? Estimate the value of pi using Monte-Carlo Simulation. Use 15 iterations to generate numbers. Compare result with its known value up to 3 decimal places.

3. What is the significance of lost calls in telephone system simulation? How do you measure the utilization and occupancy of a link in telephone system simulation? Explain.

4. The following numbers have been generated **0.54, 0.73, 0.98, 0.11 and 0.68.** Use the Kolmogorov–Smirnov (KS) test to check whether the given numbers are uniformly distributed or not at 95% confidence interval.

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5. Using the **Chi-square test** with $\alpha = 0.05$, check for the **uniformity** of the given random numbers.

0.87, 0.69, 0.12, 0.83, 0.60, 0.95, 0.01,
0.93, 0.27, 0.58, 0.23, 0.99, 0.75, 0.19,
0.28, 0.15, 0.88, 0.36, 0.89, 0.33, 0.68,
0.31, 0.35, 0.49, 0.64, 0.91, 0.05, 0.28,
0.41, 0.43

6. What is **Distributed Lag Model**? Find the growth in national consumption for **five years** using the model given. Assume the initial income $Y_{-1} = 80$ and take the government expenditure in the **5 years** to be as follows:

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Year	G
1	20
2	25
3	30
4	35
5	40

7. Explain, derive and write the **CSMP-III** program for simulation of an automobile suspension wheel.
8. Draw the **Cobweb Model (in graph)** for

   a. Fluctuation of market price.
   b. Cobweb model for market economy graph from the given data.

   $D=12.4-1.2P$

   $S=8.0-0.6P_{-1}$

   $D=S$

   $P0=1.0$

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9. Write short notes on: (Any one) 1*4 = 4

   a. Stochastic Activities
   b. Principle of Modeling
   c. Properties of Random Number

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Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor | Semester: 2nd | Year : 2026
Programme : BE Computer | Time : 3 hours | Full Marks: 100
Subject: Simulation and Modeling | **Set: A** | Pass Marks: 45

Pre-Board Examination- 2083(Spring 2026)

Candidates are required to give their answers in their own words as practicable. The figures in the margin indicate full marks.

Attempt All the questions:

1. a) What is system environment?Explain with reference to open system and close system.

OR

What is simulation and modeling? Explain the various models that are involved while simulating a system. **(7)**

b) What do you mean by time advance mechanism? Differentiate between next event and fixed increment time advance mechanism. **(8)**

2. a) Explain the structure of CSMP III. Write CSMP code for automobile suspension system. **(7)**

b) Draw the cobweb model for the market economy graph from the given data: **(8)**

$$D = 12.4 - 1.2P$$
$$S = 8.0 - 0.6P_{-1}$$
$$D = S \text{ and } P_0 = 1.0$$

OR

What is the analog method? Design an analog computer for the human liver with its differential equation.

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3. a) Explain the feedback system and interactive system simulation. **(7)**

b) Why is the Monte Carlo method best for computing a static model? Use it to solve the following integral using twenty samples:

$$\int_0^1 \frac{4}{1+x^2} dx$$ **(8)**

4. a) Compare and contrast lost call and busy call simulation of a telephone call with a suitable diagram. **(7)**

b) What is bootstrapping? How are arrivals of an event generated and analyzed in discrete system simulation? **(8)**

5. a) Define pseudo-random numbers. The following numbers have been generated: **(7)**

0.39 ,0.67 , 0.78 , 0.55

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Use the Kolmogorov–Smirnov test with $\alpha = 0.05$ to determine if the hypothesis that the numbers are uniformly distributed on the interval [0, 1] can be rejected. [Use the critical value of D for $\alpha = 0.05$ and $N = 5$ as 0.565.]

b) A sequence of random numbers is given below. Use the chi-square test with $\alpha = 0.05$ to test whether these numbers are serially auto-correlated. (Use $\chi^2_{0.05,7} = 14.067$) **(8)**

49, 95, 82, 19, 41, 31, 12, 53, 62, 40, 87, 83, 26, 1, 91, 55, 38, 75, 90, 35, 71, 57, 57, 27, 85, 52, 8, 35, 57, 88, 38, 77, 86, 29, 18, 9, 96, 58, 22, 8, 93, 85, 45, 79, 68, 20, 11, 78, 93, 21, 13, 6, 32, 63, 79, 54, 67, 35, 18, 81, 40, 62, 13, 76, 74, 76, 45, 29, 36, 80, 78, 95, 25, 52

6. a) Consider a factory that manufactures footballs, taking 20 to 40 minutes per ball. The ball is moved from the generation station to the inspection machine, taking 2 minutes. There are 3 inspection machines at one place, each needing 30 to 60 minutes for inspection, and 30% of footballs are rejected. Simulate for 1000 transactions. Draw the GPSS block diagram to simulate this system. **(7)**

b) "Initial bias decreases the efficiency of the system." Illustrate the statement with its solution. **(8)**

OR

Write SIMSCRIPT code for the main routine of a telephone call simulation.

7. Write short notes on (any two): **2×5=10**
   a) Differential equation
   b) Real-time simulation
   c) Measuring utilization and occupancy
   d) Properties of Random number
   e) SSQM

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Roshan
7 August 2026 10:46 am

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Semester: VI Spring

Level: Bachelor
Program: B.E. Computer
Course: **Simulation and Modeling**

Year : 2026
F.M. : 100
P. M. : 45
Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1 a) Define the terms System, Model, and Simulation. Explain the relationship among them with the help of a suitable example. 7

b) Draw a flowchart illustrating the Simulation Model Development Life Cycle, beginning with problem identification and ending with implementation. Briefly explain each stage shown in the diagram. 8

2 a. Given the following market equations: $D = 18 - P, S = 6 + 0.5P_{t-1}$ with an initial price of $P_0 = 3$, draw the Cobweb Model for at least four iterations and comment on whether the market converges, diverges, or oscillates. 8

b. Explain the experimental nature of simulation. Why is simulation often considered an experimental technique rather than an exact analytical solution? 7

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3 a. Explain different types of time advancement mechanism? Give examples representing the types. 7

b. Explain the concept of Hybrid Simulation. Discuss how it combines continuous and discrete-event simulation models and state its advantages over using either approach alone. 8

4 a. Discuss how feedback systems are implemented in system dynamics models explain with the help of diagram. 7

b. What are properties of Random Numbers? The following pseudo numbers are generated using a generator, make 5 intervals and determine if they are uniform using chi-square 8

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test at 5% level of significance, also state your conclusion. 0.42, 0.35, 0.21, 0.43, 0.57, 0.81, 0.68, 0.26, 0.24, 0.29, 0.82, 0.62, 0.25.

OR

c. The following 20 four-digit random numbers were generated. 123, 112, 555, 345 787, 224, 111, 678, 888, 444, 223, 998, 111, 246, 556, 909, 666, 121, 432, 999, Classify the numbers into poker categories and perform the Poker Test

5 a. A hospital receives patients according to a Poisson process with an average arrival rate of 10 patients per hour. A doctor serves patients at an average rate of 12 patients per hour. For MM1 queue, calculate: 8

i. Server utilization (ρ)
ii. Average number of patients in the system (L)
iii. Average number of patients in the queue (Lq)
iv. Average waiting time in the system (W)

b. Explain Jockeying? Does it make any change in waiting time compared to various queuing discipline. 7

OR

c. "Call gets lost when link is not available or line is busy" Verify this statement on the basis of various states involved in this simulation.

6 a. Parts are being made at the rate of one every 8 minutes. They are of two types, A and B, and are mixed randomly, with about 20% being type B. A separate inspector is assigned to examine each type of part. The inspection of A part takes 6±2 minutes and B parts take 18±10 minutes. Both inspectors reject about 10% of parts they inspect. Simulate it for 500 parts. Draw GPSS diagram and write code for same. 8 7

b. Explain organization of SIMSCRIPT with proper diagram.

7 Write short notes on (any two) 10

a. Feedback System
b. Elimination of Initial bias
c. CSMP III

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Pokhara University

Everest Engineering College

Final Internal Assessment
Spring-2026

Level: Bachelor

Program: BE CMP (6TH Sem) — **F.M. 100**
Section: A/B — **P.M. 45**
Subject: Simulation and Modeling — **Time:3hr**

Attempt all the questions.

1 a) Explain steps of simulation with help of flowchart. 8

b) How monte Carlo method is used to solving real world problems?Explain it use to determine the value of π 7

2 a) Explain the predator-prey model. Describe the impact of absence of prey for this model. 7

b) Explain Three type of statements that use CSMP III with example 8

3 a) Design analog computer for automobile suspension wheel and model of liver 8

b) Use Runs up and runs down test to determine the following sequence of number can be accepted or rejected on hypothesis of independence where $\alpha = 0.05$ 7

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49

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4 a) How do you handle lost call in telephone simulation.Explain with necessary figures. 8

b) How are utilization and occupancy measured in discrete system simulation? Explain with suitable example 7

5 a)Consider a sample of 16 data points 0.15,0.32,0.48,0.74,0.22,0.91,0.63,0.08,0.55,0.39,0.27,0.83,0.17,0.70,0.44,0.98. Perform a Kolmogorov-Smirnov(KS) Test to determine if the sample follow a uniform distribution .Use a significance level of α=0.05 7

b) Explain organization of SIMSCRIPT program with suitable diagram. 8

6 a) Write GPSS Code for manufacturing shop model when there is single inspection possible at a time. Also draw a block diagram. 8

b) How would you use replication of runs to calculate means, variances and confidence interval of output paramater? Describe. 7

7 Write short notes on: (Any two) 2*5 =10

a) Elimination of initial bias

b) Generation of Arrival Pattern

c) Real Time Simulation

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POKHARA ENGINEERING COLLEGE

Final Assessment

Level: Bachelor | Semester – Spring | Year : 2026
Programme: B.E. Computer | Full Marks: 100
Course: Simulation and Modeling | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the concept of a simulation model and discuss when simulation should and should not be used. Give suitable real-world examples. 2+3+3

   b) Explain the Monte Carlo method of simulation. Using the Monte Carlo approach, describe how the value of π can be estimated. 2+5

2. a) Explain the three general types of statements in CSMP III? Derive a CSMP III program for the automobile suspension problem. 4+4

   b) How simulation differs from analytical solution? What are the steps in simulation study? 3+4

3. a) Draw the analog computer diagram to simulate a system whose behaviour is expressed by the following equations: 8

$$dx_1/dt = -k_{12}x_1 + k_{21}x_2 - k_{31}x_3$$

$$dx_2/dt = k_{12}x_1 - (k_{21} + k_{23})x_2 + k_{24}x_3$$

$$dx_3/dt = k_{23}x_2 - k_{12}x_1$$

   b) Explain different time advance mechanisms in simulation with example. 7

4. a) Based on the runs up and runs down, determine whether the following sequence of 40 numbers is such that the hypothesis of independence can be rejected where α = 0.05. (Critical value 1.96) 8

.41 .68 .89 .94 .74 .91 .55 .62 .36 .27
.19 .72 .76 .08 .54 .02 .01 .36 .16 .28
.18 .01 .95 .69 .18 .47 .23 .32 .82 .53
.31 .42 .73 .04 .83 .45 .13 .57 .63 .29

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b) What are the various types of calls. Simulate telephone system for lost call system. 2+5

5. a) Parts are being made at rate of one every 9 ± 3 minute. As they are finished, the parts go to an inspector who tasks 15 ± 3 minute to examine each part and reject 20% of the parts. Simulate it for 500 parts. Draw GPSS diagram and write code for same. 5+3

b) What is random number? Explain the properties of random number. 7

6. a) "Replication of runs will refine simulation output". Justify this statement with statistical analysis. 8

b) Describe initial bias. How it affects the output of simulation. Explain in detail various techniques of elimination of initial bias. 7

7. Write short notes on **any two:** 2×5

a) Predatory Prey Model

b) Feedback System

c) Queue and Depart

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POKHARA UNIVERSITY
SCHOOL OF ENGINEERING

Level: Bachelor | Semester –SPRING | Year : 2026
Programme: Computer VI | Full Marks: 100
Course: Simulation & Modelling | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What are Simulation and Modelling? Explain the main components of a traffic signal simulation system with suitable assumptions. 7
   b) Mention the equilibrium point of given system also draw the cobweb model for the following equation: 8
   D=12.4-1.2P
   S=1.0+0.9P_{-1}
   P_0=1.0

2. a) How will you find the value of ∏ using Monte Carlo Method? Use 15 iteration to generate the value. Also estimate the error percentage. 7
   b) Explain the various components of Analog Computer? Draw and explain the analog computer for Human Liver system. 3+5

3. a) Why is the concept of delayed calls important in simulation? Explain how delayed calls are modelled in a telephone system simulation. 8
   b) Define Linear & partial differential equation with an example. Write CSMP III code for Wheel suspension system. 7

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4. a) Differentiate between fixed and next time advance mechanism. 7

OR

Differentiate between occupancy and utilization in system simulation. Also mention the mathematical expressions of each.

b) A random number generator produced the following sequence of observations:

15, 18, 16, 20, 19, 22, 24, 21, 23, 20, 25, 27, 26, 28, 24

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To verify whether the sequence exhibits randomness, apply the Run Test at the 5% significance level. The critical value for the standard normal distribution at the 5% significance level is $Z_{\alpha/2} = 1.96$

5. a) Employ the Linear congruential generator, to generate a sequence of 10 random numbers given r_1=987, r_2=535 and modulo m=800. Perform KS Test to check uniformity of the random under 95% confidence interval, Where D (0.05,10) = 0.410. 8

OR

Using the Middle Square Method, generate a sequence of 5 random numbers starting with the seed S_0=5731 Ensure each step uses the middle 4 digits of the squared number. Explain whether the method demonstrates randomness or suffers from any issues.

b) In a manufacturing shop, machine tools turn out parts at the rate of one every five minutes. As they are finished the posts go to inspector, who takes 5±2 minutes to examine each part and rejects 15% of parts. Each part will be represented by one transaction and time unit selected for problem will be one minute. Simulate for 90 parts to leave the system. Write a GPSS code also draw a diagram for the above scenario. 7

a) Define SIMSCRIPT program. Write Simscript program for Inventory Management System with necessary assumptions. 8

b) What is replication of run? Mention the nature of problems occurred during Simulation with explanation of each. 7

Write short notes on **any two**: 2×5

a) Elimination of initial bias

b) Predator Prey Model with necessary equations

c) Feedback System

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Best of luck

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GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Level: Bachelor
Programme: BE CE VI
Course: **Simulation and Modelling**

Semester: Spring

Year : 2026
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the principle used in Modeling. Define entity, attribute and activities of a system with example. 8
   b) A bank receives 12 customers/hour. The cashier serves 15 customers/hour. Given: $\lambda=12$ and $\mu=15$. Calculate utilization, average customer in system and average waiting time in queue. 7

2. a) Use Monte Carlo method to calculate : $I = \int_2^6 3x^2 dx$ 7
   b) Draw the Cobweb model for market economy graph from given data. 8

$P_0=1.0$, a=12.4, b=1.2, c=1.2, d=0.9

3. a) Construct analog computer for Electric circuit with differential equation. 7
   b) Explain the Hybrid Computers and Digital analog simulators. 8

4. a) Explain the process for the delay-call system in a simulation of a Telephone system. 8
   b) Explain counter and summary statistics and the measuring utilization and occupancy. 7

5. a) Show all the random number Generations methods with examples. 8

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b) Consider sequence of ~~30~~ 25 numbers as given below. Test whether the 3rd, 8th, 13th, 18th, 23rd, ~~28th~~, number in the sequence are autocorreleted with $\alpha=0.05$, and $Z_{0.025}=1.96$. The numbers are: 7

0.12	0.01	0.23	0.28	0.89
0.64	0.28	0.83	0.93	0.99
0.33	0.35	0.91	0.41	0.60
0.75	0.88	0.68	0.49	0.05
0.95	0.58	0.19	0.36	0.69

6 a) A Car arrives in every 10 minutes, if there is queue then stays in queue to fill up the petrol in 5 to15 minutes and go to parking. A car park for 20 minutes in parking area for lunch and after that leaves the parking. Draw BLOCK diagram and GPSS program. 8

b) Explain Elimination of Initial Bias for output analysis. 7

7. Write short notes on **any two:** 2*5

i) Verification and validation of Model

ii) Time Advancement Mechanism

iii) Write a SIMSCRIPT program for PREAMBLE of telephone system.

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Pokhara University
Lumbini Engineering College
Pre-Board Examination-2083

Level: Bachelor
Program: BE (Computer) 6th sem
Subject: Simulation and Modeling

Full Marks: 100
Pass Marks: 45
Time: 3 hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt all Questions.

1) a) Define simulation and explain the importance of simulation with its application. [8]
   b) Why do we need to simulate a system? Define system and its elements with example. [7]
2) a) Define endogenous and exogenous activities. [8]
   b) Why do we need the Monte-Carlo method of simulation? Solve $\int_0^3 \sqrt{1+x^2}dx$ using ten samples. [7]
3) a) Differentiate between continuous and discrete system. Design an analog computer for automobile suspension wheel. [8]
   b) Explain the term verification, validation and calibration of models. [7]
4) a) What is Distributed Lag Model? Find the growth in national consumption for five years using the model given. Assume the initial income Y_{-1} is 80 and take the government expenditure in the 5 years to be as follows. [8]

Year	G
1	20
2	25
3	30
4	35
5	40

   b) Define the types of statements in CSMP III. Write CSMP III program for Autopilot system. [7]
5) a) What are the various types of calls? Simulate telephone system for lost call system. [8]

b) Based on runs up and runs down, determine whether the following sequence of 40 numbers is such that the hypothesis of independence can be rejected or accepted. [7]
Significance Level: $\alpha = 0.05$ (two-tailed test). Critical Values; ± 1.96

0.29	0.63	0.57	0.13	0.45	0.83	0.04	0.73	0.42	0.31
0.53	0.82	0.32	0.23	0.47	0.18	0.69	0.95	0.01	0.18
0.28	0.16	0.36	0.01	0.02	0.54	0.08	0.75	0.72	0.19
0.27	0.36	0.62	0.55	0.91	0.74	0.94	0.89	0.68	0.45

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6) a) Workers come to a supply store at a rate of one every 5±2 min. Their requisitions are processed by one of two clerks who take 8 ± 4 min for each requisition. The requisitions are then passed to a single storekeeper who fills them one at a time taking 4±2 min for each request. Draw the GPSS block diagram to simulate the queue of workers and measure the distribution of time taken for 1000 requisitions to be filled. [8]

b) What is SIMSCRIPT? Explain the Organization of SIMSCRIPT program with necessary diagram. [7]

7) Write short notes on (Any two) [2× 5]

a) Utilization and occupancy

b) Pseudo random number

c) Predator pray model

The End

COMPILED BY

NIRAJ BHATTA

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UNITED TECHNICAL COLLEGE

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE | Full Marks : 100
Course: Simulation and Modeling | Pass Marks : 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1 a. What do you mean by simulation and modeling? Explain the steps involved in a simulation study with the help of a flow chart. 7

b. Draw Cobweb model for market economy graph from the given data: 8

- $D = 12.4 - 1.2P$
- $S = 7.0 + 0.6P_{-1}$
- $D = S$
- $P_0 = 1.0$

2 a. Find the value of PI using the Monte Carlo method and find out its deviation from standard value. 7

b. What is differential equation? Derive differential equation for mechanical system of suspension wheel and design an analog computer for it. 8

3 a. A sequence of 1000 four-digit numbers has been generated, and an analysis indicates the following combination and frequencies: 8

Combination Distribution (i)	Observed frequency (Oi)
4 different digits	540
3 like digits	50
4 like digits	20
1 pair	320
2 pairs	70

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Use Poker's test to determine if these random numbers are independent. $X_{0.05,4}=9.49$

b. Explain, derive and write the CSMP-III program for simulation of an autopilot. 7

4 a. What are the various types of calls? Simulate telephone system for lost call system. 7

b. What is discrete system? How arrival patterns can be generated for these types of system. 8

5 a. The following numbers have been generated 0.39, 0.67, 0.78, and 0.55. Use the Kolmogorov-Smirnov Test to check whether given numbers are uniformly distributed or not. (Use the critical value of D for alpha=0.05 and N=5 is 0.565.) 7

b. The two Digit random numbers generated by a multiplicative congruential method are given below. Test these data for uniform distribution using Chi-square. Is it acceptable at 95% confidence level? (Use x^2 0.05,9=16.9) 8

36, 91, 51, 02, 54, 06, 58,06, 58, 02, 54, 01, 48, 97, 43, 22,83, 25, 79, 95, 42, 87, 73, 17, 02,42, 95, 38, 79, 29, 65, 09, 55, 97, 39, 83, 31, 77, 17, 62, 03, 49, 90, 37, 13, 17, 58, 11, 51, 92, 33, 78,21, 66, 09, 54, 49, 90, 35, 84, 26, 74, 22, 62, 12, 90, 36, 83, 32, 73, 31, 04, 34, 71, 40, 67, 58, 05, 56, 22, 58, 77, 71, 10, 73, 23, 57, 13, 36, 89, 22, 68, 02, 44, 99,27, 81,26,85,22

6 a. Define SIMSCRIPT program. Explain organization of SIMSCRIPT statement. 7

b. Discuss the role of replication of runs in analysis of output for simulation and modelling. 8

7 Attempt Any two (5×2=10)

a. Predator prey model

b. Utilization and occupancy

c. Pseudo random number

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